fMRI prediction models — SOTA review

2026-06-14

SOTA models

1. Brain-JEPA — NeurIPS 2024 Spotlight

arXiv: 2409.19407 **Status**: Published — NeurIPS 2024 Spotlight (top 3% of submissions). **Architecture**: JEPA (Joint Embedding Predictive Architecture) + ViT on **ROI time-series** (450 ROI × 160 timesteps). The predictor reconstructs the **embeddings** of target blocks instead of raw voxels (LeCun's paradigm, adapted from I-JEPA). **Pretraining dataset**: UK Biobank — 32,130 subjects (80% of 40,162 available), TR 0.735s.

Evaluation dataset	Task	Acc	F1
UK Biobank	Sex	**88.17**	**88.58**
HCP-Aging	Sex	**81.52**	**84.26**
ADNI (n=189)	NC vs MCI	**76.84**	**86.32**
ADNI	Amyloid β+/-	71.00	75.97
MACC	NC vs MCI (Asian cohort)	65.98	64.67
OASIS-3	AD Conversion	69.00	67.32
CamCAN	Depression	72.73	67.45

Downstream protocol: fixed 6:2:2 train/val/test split across all datasets, results averaged over 5 independent runs. Subject-awareness not explicitly stated.

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2. BrainLM — ICLR 2024 Spotlight

arXiv: biorxiv 2023.09.12.557460 **Status**: Published — ICLR 2024 Spotlight. **Architecture**: Vision Transformer in Masked Autoencoder (MAE) mode on ROI time-series. Masks 75% of patches, reconstructs **raw values**. 111M parameters. **Pretraining datasets**: UK Biobank (76,296 recordings, ~6,450 hours) + HCP (1,002 recordings, ~250 hours) = 77,298 recordings / 6,700 hours. Atlas: AAL-424 ROI.

Evaluation dataset	Task	MSE (z-scored)
UK Biobank	Age (regression)	**0.503 ± 0.021**
UK Biobank	PTSD (PCL-5)	0.015 ± 0.0003
UK Biobank	Anxiety (GAD-7)	0.073 ± 0.003
UK Biobank	Neuroticism	0.069 ± 0.004

No Sex / MCI / AD / MMSE numbers reported. Only z-scored MSE regression on UK Biobank held-out.

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3. SLIM-Brain — preprint (Dec 2025, under OpenReview review)

arXiv: 2512.21881 **Status**: Not yet published — preprint, under OpenReview review (anonymous submission). Self-reported SOTA, not peer-reviewed. **Architecture**: **4D Hiera-JEPA** — Hiera (hierarchical pyramidal ViT, multi-stage) combined with the JEPA paradigm. Input = **4D volumes** `(T, X, Y, Z)`. 4D tubelets, global attention within each stage. **Pretraining datasets**: HCP + CHCP + AOMIC PIOP1 + AOMIC PIOP2 + ABCD = **4,129 sessions** combined (70% of total). Harmonized to 2mm isotropic / TR 0.72s.

Evaluation dataset	Task	Acc	F1 / other
HCP	Sex	**91.1**	F1 **91.1**
HCP	Fingerprint	98.5	F1 98.1
ADNI	MCI classification	**69.12**	F1 68.96
ADHD-200	ADHD	63.53	—
PPMI	Parkinson	70.40	—
ABIDE	Age (classification)	64.41	—
ABIDE	Age (regression)	—	MSE 0.2175

Beats Brain-JEPA on HCP Sex (87.1) and ADNI MCI (64.53). Downstream protocol: 70/10/20 train/val/test split, 3 independent runs. Subject-awareness not explicitly stated.

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4. BrainGFM — preprint (June 2025)

arXiv: 2506.02044 **Status**: Not published — preprint, no confirmed venue. **Architecture**: **Graph Foundation Model** on connectomes. Combines **Graph Contrastive Learning + Graph Masked Autoencoder** + meta-learning + graph prompts + language prompts. Multi-atlas (8 parcellations). **Pretraining datasets**: 27 datasets covering 25 disorders, 25,000+ subjects / 60,000 scans / 400,000 graph samples. Identified examples: OpenNeuro, UK Biobank, HCP, ABIDE, ABIDE II, ADHD-200, OASIS, ADNI 2, HBN, SubMex_CUD, UCLA_CNP (full list in Appendix N).

Evaluation dataset	Task	AUC	Acc
ADHD-200	ADHD	70.6	72.2
ABIDE II	Autism (ASD)	71.2	73.5
ADNI 2	**Alzheimer (AD)**	**80.3**	**85.1**
HBN	Major Depression	83.6	85.5
HBN	Anxiety	85.2	86.3
HBN	OCD	80.4	85.8
HBN	PTSD	83.2	86.3
SubMex_CUD	Cocaine Use	71.1	74.6
UCLA_CNP	Schizophrenia	84.2	86.7
UCLA_CNP	Bipolar	73.5	76.3

No Sex / Age / Parkinson results reported. Schaefer-100 atlas used.

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5. LCM (Large Connectome Model) — AAAI 2026

arXiv: 2510.18910 **Status**: Published — AAAI 2026 (accepted). **Architecture**: **Decoder-only Transformer** (GPT-style) on connectomes (FC matrix). MHSA + MHCA between connectome features and "brain-environment-interaction" (BEI) tokens (encoding covariates: sex, age, scanner, site). **Pretraining datasets**: HCPA (713 subj / 4,863 scans) + HCPYA (248 / 3,293) + ADNI (138 / 138) + PPMI (209 / 209) + ABIDE (1,025 / 1,025) + Taowu (40 / 40) + Neurocon (41 / 41) = ~10,036 scans / 7 datasets.

Evaluation dataset	Task	F1
HCP-Aging	Sex	**73.94 ± 2.45**
HCP-YA	Sex	**72.23 ± 1.92**
ABIDE	Sex	87.34 ± 4.48
ADNI	**Alzheimer**	**85.33 ± 7.35**
PPMI	Parkinson	84.18 ± 11.63
ABIDE	Autism	72.50 ± 1.91

Downstream protocol: subject-aware 5-fold CV for HCPA/HCPYA/ADNI, 10-fold for others. Pretraining and finetuning always from the same CV fold's training set to prevent leakage.

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6. BNT (Brain Network Transformer) — NeurIPS 2022

arXiv: 2210.06681 **Status**: Published — NeurIPS 2022. **Architecture**: Pure Transformer on connectome graphs (not a GNN). Each ROI = one token, node features = connection profile. Key innovation = **Orthonormal Clustering Readout** (pooling via K learned orthogonal prototypes). **Pretraining**: None — supervised end-to-end.

Evaluation dataset	Task	Metric	Score
ABIDE	Autism	AUROC	**80.2%**
ABCD	Sex (n=7,901)	—	not extracted
ADNI	NC vs MCI	Acc	**78.90**

BNT beats Brain-JEPA on ADNI NC/MCI (78.90 vs 76.84) — reported as comparator in Brain-JEPA paper.

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7. OViTAD — Brain Sciences 2023

arXiv: biorxiv 2021.11.27.470184 **Status**: Published — Brain Sciences 2023 (MDPI journal, IF ~3, not a top venue). **Architecture**: Standard **ViT-B/16** on individual **2D fMRI slices** (no 3D, no time). Subject-level inference via **majority vote**. No architectural innovation — only hyperparameter tuning. **Pretraining**: None — supervised end-to-end on ADNI.

Evaluation dataset	Task	F1
ADNI (n=284)	AD vs HC	**0.99 ± 0.02**
ADNI (n=284)	HC vs MCI	**0.97 ± 0.03**

Test set = only 31 subjects. σ=0.02 on n=31 is noise-dominated. Subject-aware 80/10/10 split (226 train / 27 val / 31 test).

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8. BrainNetCNN — NeuroImage 2017

arXiv: Kawahara 2017 PDF **Status**: Published — NeuroImage 2017 (Elsevier, IF ~5). Historical baseline reused by modern connectome papers. **Architecture**: CNN with **3 custom topology-aware filters** on connectivity matrix — Edge-to-Edge (E2E, aggregates row+column per cell), Edge-to-Node (E2N, aggregates per ROI), Node-to-Graph (N2G, global pooling). **Pretraining**: None — supervised end-to-end.

Evaluation dataset	Task	Score
Preterm DTI (original)	Bayley-III scores	**not comparable**

Original paper used DTI (diffusion MRI, not fMRI) on preterm infants. Architecture reused as supervised baseline by BNT / BrainGFM / BrainGB on adult fMRI.

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9. SwiFT — NeurIPS 2023

arXiv: 2307.05916 **Status**: Published — NeurIPS 2023. **Architecture**: **4D Swin Transformer** on raw fMRI volumes (no atlas). 4D windows `(X, Y, Z, T)` + windowed self-attention + shifted windows between layers + hierarchical (4 stages). Closest volumetric Transformer paradigm. **Pretraining**: Light contrastive SSL (composition not specified in main paper).

Evaluation dataset	Task	Acc	Other
OASIS-3	AD Conversion	65.00	F1 66.80
ADNI	MCI	64.45	—
ABIDE	Age (cls)	62.22	MSE 0.4137
ADHD-200	ADHD	60.81	—
PPMI	Parkinson	58.10	—
HCP / ABCD / UKB	Sex, Age	—	not extracted

Numbers reported as comparator within Brain-JEPA Table 9 and SLIM-Brain Table 1 — not directly extracted from the SwiFT paper itself.

Datasets — dimensions and access

Dataset	# subjects	TR (s)	Typical T	Access mode	Used in pretraining by	Used in evaluation by
UK Biobank	40,000+	0.735	490	Paid DUA (~£500-1000), 2-6 months	Brain-JEPA, BrainLM, BrainGFM	Brain-JEPA, BrainLM, SwiFT
HCP-YA	1,200	0.72	1,200	Open with DUA, fast	BrainLM, SwiFT, LCM	LCM, SLIM-Brain (Sex)
HCP-Aging	700	0.8	~470	NDA controlled, 1-3 months	Brain-JEPA (eval only), LCM	Brain-JEPA, LCM
CHCP	~370	0.72	varies	Chinese open access	SLIM-Brain	—
AOMIC PIOP1	216	2.0	~480	Open access	SLIM-Brain	—
AOMIC PIOP2	226	2.0	~480	Open access	SLIM-Brain	—
ABCD	11,800+	0.8	~375	NDA controlled, 1-3 months	SLIM-Brain, SwiFT, BrainGFM	SwiFT
ADNI / ADNI 2	~1,000	3.0	~140	DUA, 1-2 weeks	LCM (138 subj only)	Brain-JEPA, BrainGFM, LCM, SLIM-Brain, OViTAD

Dataset	# subjects	TR (s)	Typical T	Access mode	Used in pretraining by	Used in evaluation by
ABIDE I	1,112	1.5-3	76-300	Open (NITRC / Preprocessed Connectomes Project)	BrainGFM	BNT, Brain-JEPA, BrainGFM, SLIM-Brain, LCM
ABIDE II	1,114	varies	varies	Open	BrainGFM	BrainGFM
ADHD-200	776	1.5-2.5	70-280	Open (NITRC)	BrainGFM	BrainGFM, SLIM-Brain
PPMI	~500	2.4	~200	DUA, 1-2 weeks	LCM	LCM, BrainGFM, SLIM-Brain
HBN	3,000+	0.8	varies	Open (AWS S3)	BrainGFM	BrainGFM
OASIS-3	1,098	2.2	~180	Registration + proposal, 1-7 days	—	Brain-JEPA (AD Conversion)
MACC	restricted	varies	varies	Restricted (Asian collaborators only)	—	Brain-JEPA (NC/MCI Asian)
CamCAN	700	1.97	261	Open with registration	—	Brain-JEPA (Depression)
SubMex_CUD	restricted	varies	varies	Restricted	BrainGFM	BrainGFM (Cocaine)
UCLA_CNP	272	2.0	152	Open (OpenNeuro)	BrainGFM	BrainGFM (Schizo, Bipolar)
Taowu	40	2.0	~120	Open (OpenNeuro)	LCM	LCM (Parkinson)
Neurocon	41	2.0	~140	Open (OpenNeuro)	LCM	LCM (Parkinson)

Access categories: - **Open access** (immediate download via Python libraries like `nilearn`): ABIDE I/II, ADHD-200, AOMIC PIOP1/PIOP2, CHCP, HBN, UCLA_CNP, Taowu, Neurocon - **Registration + short proposal** (1-7 days): OASIS-3, CamCAN - **DUA, fast** (1-2 weeks): ADNI, PPMI, HCP-YA - **NDA controlled** (1-3 months): HCP-Aging, ABCD - **Paid DUA, slow** (2-6 months): UK Biobank (~£500-1000) - **Restricted to specific collaborators**: MACC, SubMex_CUD

Cross-reference tables

Datasets used for pretraining vs evaluation

Model	Pretraining datasets	# pretrain sessions	Evaluation datasets
Brain-JEPA	UK Biobank	~32,130 subjects	UKB, HCP-Aging, ADNI, MACC, OASIS-3, CamCAN
BrainLM	UK Biobank + HCP	77,298 recordings	UK Biobank only
SLIM-Brain	HCP + CHCP + AOMIC PIOP1/PIOP2 + ABCD	4,129 sessions	HCP, ADNI, ADHD-200, PPMI, ABIDE
BrainGFM	27 datasets (HCP, UKB, ABIDE, ADHD-200, OASIS, ADNI 2, HBN, SubMex_CUD, UCLA_CNP, +18 in App N)	~60,000 scans	ADNI 2, ABIDE II, ADHD-200, HBN, SubMex_CUD, UCLA_CNP
LCM	HCPA + HCPYA + ADNI + PPMI + ABIDE + Taowu + Neurocon	~10,036 scans	HCPA, HCPYA, ABIDE, ADNI, PPMI
BNT	None (end-to-end supervised)	—	ABIDE, ABCD

Model	Pretraining datasets	# pretrain sessions	Evaluation datasets
OViTAD	None (end-to-end supervised)	—	ADNI (284 subjects)
BrainNetCNN	None (end-to-end supervised)	—	Preterm DTI (original paper)
SwiFT	Light contrastive SSL (corpus unspecified)	—	HCP, ABCD, UK Biobank

Labels used per model

Model	Label types
Brain-JEPA	DX (NL/MCI clinical diagnosis), Amyloid β +/-, Sex, Depression
BrainLM	Age (regression), PTSD, Anxiety, Neuroticism (all z-scored MSE)
SLIM-Brain	DX (MCI classification), Sex, Fingerprint, ADHD diagnosis, Parkinson, Age
BrainGFM	10 disorders: ADHD, ASD, AD (DX), MDD, Anxiety, OCD, PTSD, Cocaine, Schizo, Bipolar
LCM	Sex, DX (AD, PD, ASD, SZ)
BNT	DX (Autism), Sex
OViTAD	DX (AD vs HC, HC vs MCI)
BrainNetCNN	Bayley-III scores (preterm)
SwiFT	Sex, Age, Cognitive intelligence

Notes on metric comparability

- **Most SOTA papers report Accuracy (%) and F1 (%)**, not AUC. Direct AUC-to-Acc conversion is approximate.
- **Brain-JEPA / BrainGFM / LCM use DX clinical diagnosis labels** (NL / MCI / AD from ADNI metadata). Direct comparison to CDR-based binarisations requires label alignment.
- **No SOTA paper reports horizon-specific cognitive decline forecasting** (1Y / 2Y / 3Y splits). The closest comparator is Brain-JEPA's OASIS-3 AD Conversion (single horizon, Acc 69.00 / F1 67.32).
- **Subject-level splits**: Brain-JEPA (6:2:2), SLIM-Brain (70:10:20), LCM (subject-aware 5/10-fold), OViTAD (80:10:10 at participant level). Subject-awareness explicitly confirmed only for LCM and OViTAD.

Sources

1. **Brain-JEPA: Brain Dynamics Foundation Model with Gradient Positioning and Spatiotemporal Masking** — Dong & Li et al., NeurIPS 2024 Spotlight — arxiv.org/abs/2409.19407
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3. **SLIM-Brain: A Data- and Training-Efficient Foundation Model for fMRI Data Analysis** — Anonymous, arXiv Dec 2025 — arxiv.org/abs/2512.21881
4. **Brain Graph Foundation Model (BrainGFM): Pre-Training and Prompt-Tuning across Broad Atlases and Disorders** — arXiv 2025 — arxiv.org/abs/2506.02044
5. **Large Connectome Model (LCM): A Decoder-Only Foundation Model for fMRI** — AAAI 2026 — arxiv.org/abs/2510.18910
6. **Brain Network Transformer (BNT)** — Kan, Cui et al., NeurIPS 2022 — arxiv.org/abs/2210.06681
7. **OViTAD: Optimized Vision Transformer for the diagnosis of Alzheimer's disease** — Sarraf et al., Brain Sciences 2023 — biorxiv 10.1101/2021.11.27.470184
8. **BrainNetCNN: Convolutional neural networks for brain networks** — Kawahara et al., NeuroImage 2017 — PDF
9. **SwiFT: Swin 4D fMRI Transformer** — Kim et al., NeurIPS 2023 — arxiv.org/abs/2307.05916